

Study of a hybrid renewable energy system for a rural school in Tagzirt, Morocco

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Abstract:

The continuous rise of energy demand worldwide combined with the decrease of natural resources such as fossil fuel represents a huge energy problem which facing humanity. Industry as well as consumers must rethink how to produce and use energy at low cost price. Renewable Energy (RE) applications and energy savings are keys to meet this challenge in a sustainable way and reduce the percentage of carbon dioxide (CO₂) emission to environment, which reduces the global warming. This paper presents a methodology to systematically formulate a hybrid renewable energy system (HRES) for a rural school based at Tagzirt, Morocco, which consists of solar, wind and diesel generator as a backup resource as well as battery storage, from the preliminary design stage to the optimal operation. The computation software used for this work is HOMER Pro 3.2.3. The analysis of such hybrid energy systems configuration feeding AC school load of 23 kWh/day energy consumption with a 6.02 kW maximum load demand. The simulation results show that optimized size of components, Wind turbine (2 kW) - solar (5 kW) - diesel generator (7.8 kW) and optimized cost of energy about 1.12 \$/kWh.

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SECTION I. Introduction

In the last few years, renewable energy systems have attracted the great interest because conventional sources of energy are limited and a number of problems associated with their use, like environment pollution, large grid requirements etc. Governments of the whole world are forced for the alternative energy sources such as wind energy, solar energy and small hydro-electric power, etc. [1]

Morocco, as well as globally, being one of largest energy importer in MENA, is making concerted efforts to reduce its reliance on imported fossil fuels. Morocco has recently focused on renewable energies to reduce its energy dependence covered in majority by imports. In 2012, Morocco spent around US\$10 billion on all energy imports (crude oil and oil products, coal, natural gas and electricity). Annual electricity consumption in Morocco was 33.5 TWh in 2014, and is steadily increasing at a rate of around 7 percent each year [2], [3].

Morocco enjoys a rich potential in renewable energies mainly solar and wind. On a global scale, Morocco occupies the 9th position for solar, and the 31st for wind.

Morocco has an intensive solar radiation with an average of 5.3 kWh/m² under annual sunshine durations ranging from 2700 H in the north to approximately 3500 H in the south. For wind energy, the country

possesses a huge potential, especially along its 3500 Km coastal areas with wind speeds ranging from 7 m/s and 11 m/s [4], [5].

Taking profit of its huge potential renewable energies, Morocco has introduced an ambitious renewable energy program for the 2010-2020 period. This program is expected to generate a structural change in the energy supply via the extension of renewable energies as power generation technologies. In fact, Morocco aims to increase the installed electrical capacity based on renewable energies to 42% by 2020 through the planning of numerous projects in solar, wind and hydropower options.

A hybrid energy system incorporates two or more electricity generation options based either on renewable energy unit or fossil fuel based unit like diesel-electric generator or a small gas-turbine along with energy storage and electronic devices [6]– [9]. To accurately size the individual components of the system, simulation of the system under real operating condition like appropriate weather, insolation, wind speed and loads is necessary. The components and subsystems of a HRES are interconnected to optimize the whole system. The design of a hybrid system will depend on several requirements like location, standalone or grid tied DC or AC load, etc [10], [11]. The main objective in case of HRES is to utilize maximum proportion of renewable energy, other factors includes the financial investment, reliability and durability. The aim of this study is to investigate the level of awareness and prevailing attitudes of school students toward renewable energy (RE) in a country which is heavily reliant on fossil fuels despite having high RE resources.

SECTION II. Methodology

The school we want to equip with the multi-source energy system is located in Tagzirt in the region of Beni Mellal, Morocco. It is a rural area at the foot of Mount Tassimet (Atlas Mountains). [Table 1](#) gives more information about the location.

Table I. Geographical coordinates of the school

Latitude (°N)	Longitude (°E)	Altitude (m)	Time
32°20'14"	6°20'59"	538 m	GMT

The school contains five classrooms, an administration and a W.C. In addition, the school incorporates two functional houses. [Table 2](#) provides a list of electrical equipments present in the school alongside with their power. The simulation of the electrical consumption behavior of the school takes into account hours of work in the day, weekends, holidays and annual vacation (see Fig.1).

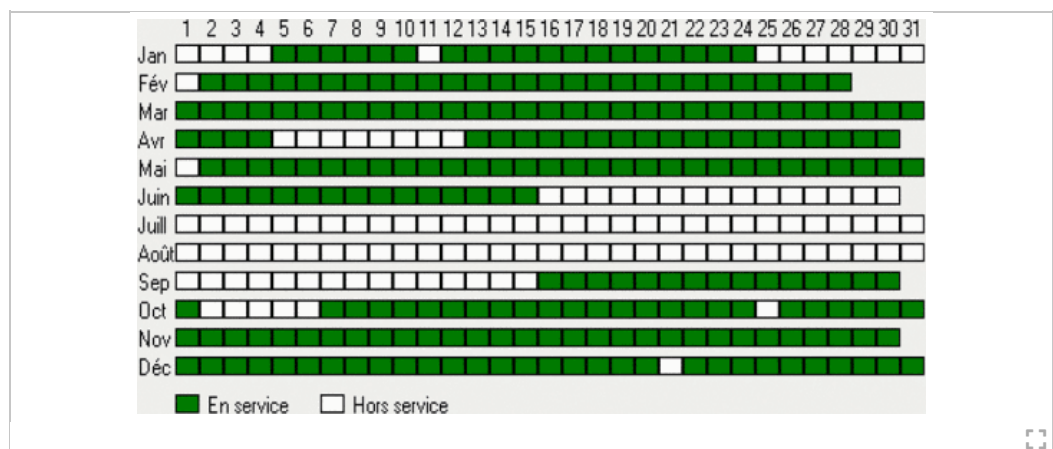


Figure 1.
Period of service during the year of the school.

Table II. Assessment of the electricity consumption of the school

Place	Electrical equipment	Power (W)	Number	Total power (W)
Classrooms (5)	Light bulb	75	20 (4 per classroom)	1500
Administration	Light bulb	75	4	300
	Computer	150	1	150
W.C.	Light bulb	75	4	300
Functional houses (2)	Light bulb	75	10	750
	TV station	80	2	160
	Refrigerator	160	2	320
	Radio	60	2	120
	Computer	150	2	300
Total electricity power				3900 W

The average daily load of the school is shown below in “[Fig. 2](#)”.

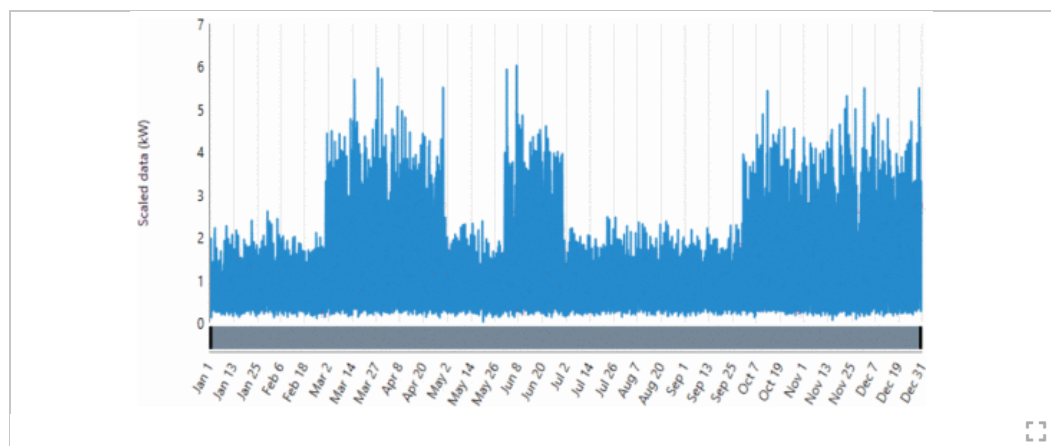
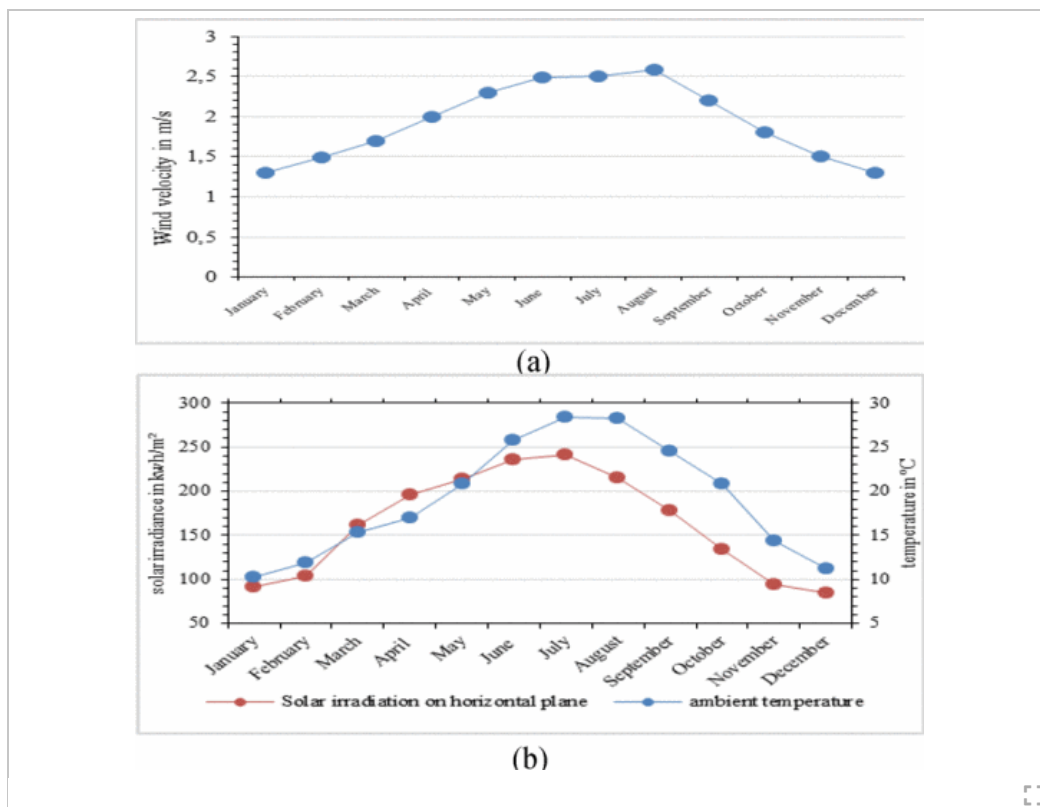


Figure 2.
Annual load profile of the school.

A. Renewable Energy Resources for Hybrid System

This work consists, mainly, on exploiting wind and solar energies to meet the energy demand of the school. [Fig. 3](#) shows the renewable energy potential of the considered location given by Meteonorm software. The annual average of wind velocity equals 1.93 m/s at 10 meters of height. Even if it is not too strong, it is sufficient for our purpose. The annual variation of wind velocity shows a pic in August equals 2.59 m/s. In fact, in this location, wind velocity is the most important during the summer and it is the less strong during the winter (see [Fig. 3-a](#)). As it is depicted in Fig 3-b, a horizontal plane in this location receives 1950 kWh/m² of solar irradiance in a year with a pic in the month of July equals 141 kWh/m². Furthermore, [figure 3-b](#) gives the annual irradiance with a temperature since it is an important parameter influencing electricity generation of PV modules. The annual average of temperature equals 19.1 °C with a pic equals 28.45 °C in the month of July. The minimum temperature recorded in this location equals 10.18 °C in January.

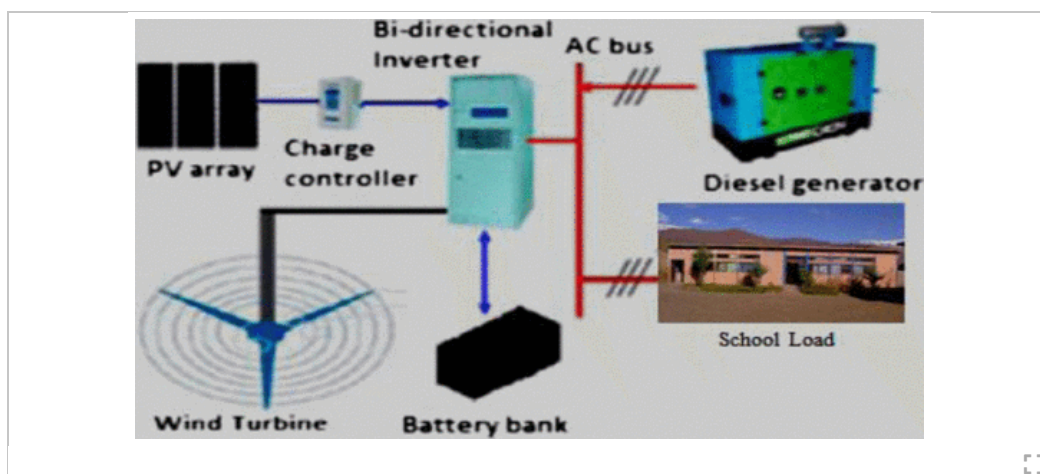
**Figure 3.**

Renewable energy potential of the considered location: (a) Wind velocity (right), (b) solar irradiation and temperature (left).

B. Hybrid System Configuration

The hybrid system consists of the following components: PV array, wind turbine, back-up AC diesel generator with battery storage and power electronics unit connected to the school site (see Fig. 4). The use of a battery bank in this system is for storing excess electricity that is generated from the wind turbine and solar panels.

To meet the energy demand of the school we studied three scenarios, all of them contain at least one renewable energy sources. The proposed energy sources are PV modules, wind turbine, batteries and, a backup diesel engine. Table 3 describes in detail the scenarios.

**Figure 4.**

Pv/wind/diesel hybrid system configuration.

Table III. Different scenarios of the proposed study

	PV modules	Wind turbine	Battery bank	diesel GENERATOR
Scenario1	*	-	*	*
Scenario 2	-	*	*	*
Scenario 3	*	*	*	*

The idea behind establishing these three scenarios is to get to the most benefic combination of energy sources regarding covering energy demand, economic cost effectiveness and, intermittence of wind and solar energy.

SECTION III.

Results and Discussion

A. Homer Pro Simulation Model

The optimal sizing and costing of the components of the proposed hybrid energy system has been using HOMER pro software (version 3.2.3). The name HOMER is an abbreviation of a Hybrid Optimization Model for Electrical Renewable and it is developed by U.S. National Renewable Energy Laboratory (NREL) [12], [13]. HOMER Pro allows simulation of grid-connected and off-grid systems which generate electricity from various combinations of solar PV modules, wind turbines, biomass based power generators, micro-turbines, fuel cells, batteries, hydrogen storage and auxiliary generators with various fuels options and different type of loads.

The simulation model has been designed HOMER Pro software, and consists of a PV system, Wind turbine system, diesel generator and battery. The schematic of this hybrid energy system model is shown in Fig.5.

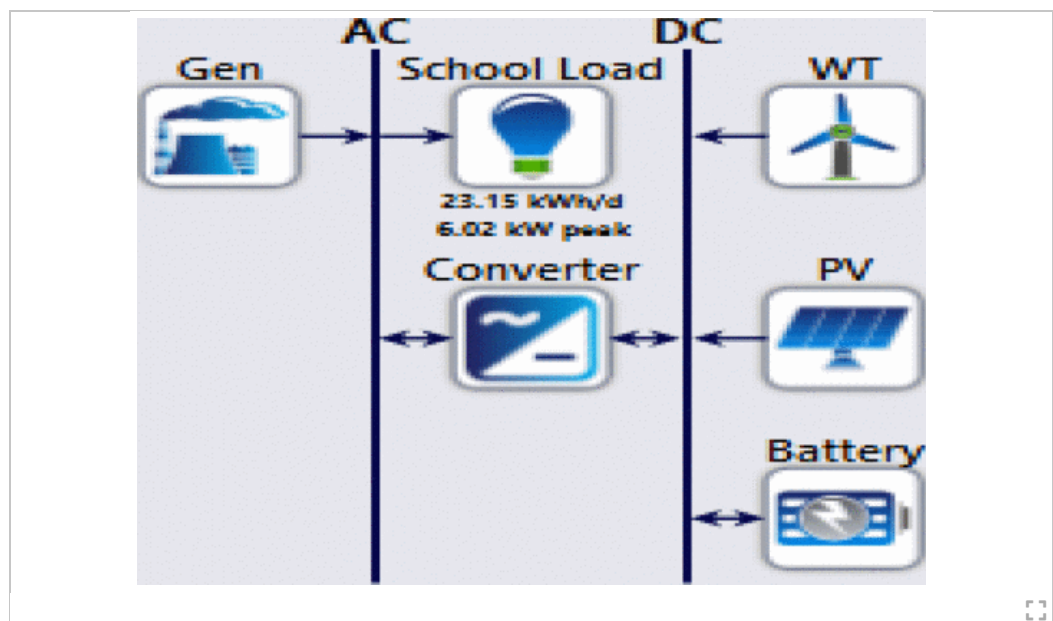


Figure 5.
HOMER simulation model of hybrid energy system.

The overall results of hybrid energy system HOMER Pro simulation model are shown in Fig.6.

RESULTS

Sensitivity Cases Left Click on a sensitivity case to see its Optimization Results.

Architecture			Cost			System		Gen							
 PV (kW)	 WT (kW)	 Gen (kW)	 LPF	 Converter (kW)	 Dispatch	 COE (\$)	 NPC (\$)	 Operating cost (\$)	 Initial capital (\$)	 Ren Frac (%)	 Hours	 Production (kWh)	 Fuel (\$)	 OBM Cost (\$)	 Fuel Cost (\$)
 6.00		7.80	40	3.00	CC	\$1.12	\$122,199	\$5,140	\$55,750	52	940	4,054	2,251	1,128	2,251
 4.00	3	7.80	40	4.00	CC	\$0.986	\$107,720	\$4,140	\$54,200	67	631	2,811	1,539	757	1,539
 2.00	4	7.80	32	4.00	CC	\$0.832	\$90,879	\$3,755	\$42,340	71	582	2,432	1,369	698	1,369
 5		7.80	40	4.00	CC	\$0.684	\$74,690	\$3,148	\$34,000	78	436	1,840	1,031	523	1,031
 4		7.80	32	4.00	LF	\$0.565	\$61,682	\$2,579	\$28,340	91	452	793	712	542	712
 6.00		7.80	40	3.00	CC	\$1.12	\$122,199	\$5,140	\$55,750	52	940	4,054	2,251	1,128	2,251
 4.00	3	7.80	40	4.00	CC	\$0.986	\$107,720	\$4,140	\$54,200	67	631	2,811	1,539	757	1,539
 2.00	4	7.80	32	4.00	CC	\$0.832	\$90,879	\$3,755	\$42,340	71	582	2,432	1,369	698	1,369

Figure 6.
Sensitivity results from HOMER software.

The results of the three different scenarios:

• *Scenario1* System architecture PV /Generator/ Battery

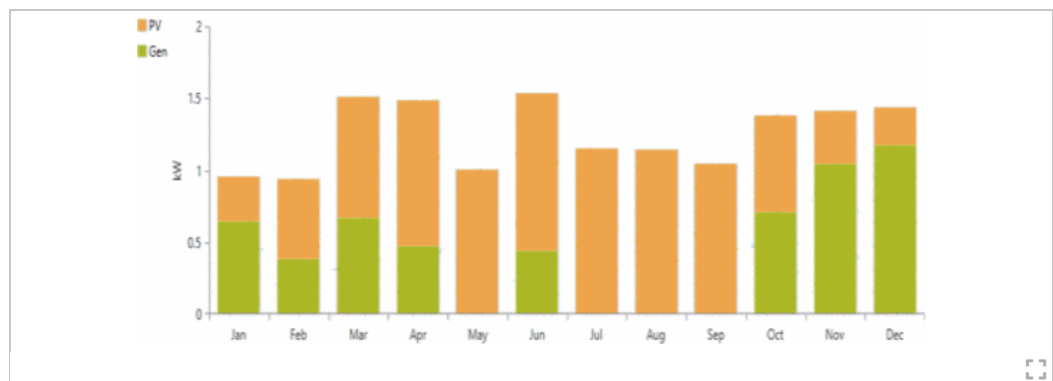


Figure 7.
Monthly average electric production (pv/generator).

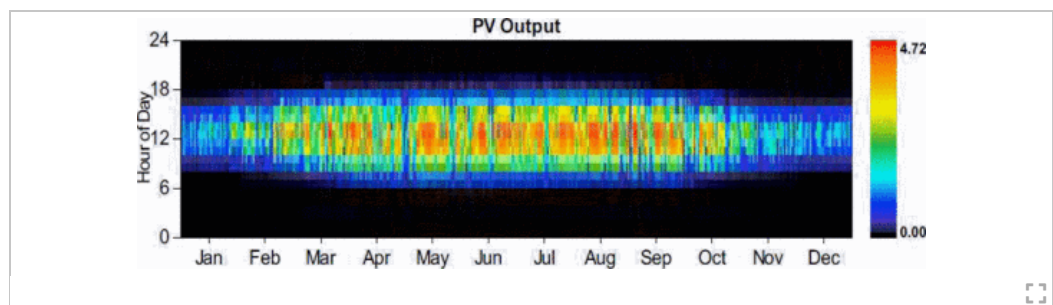


Figure 8.
PV power output (kW).

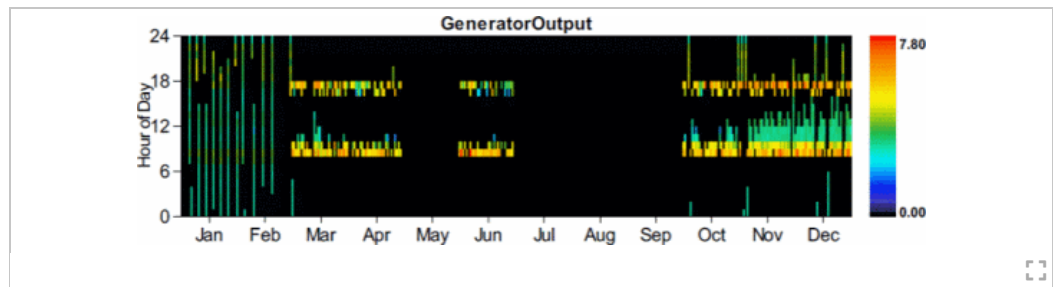


Figure 9.
Generator power output (kw).

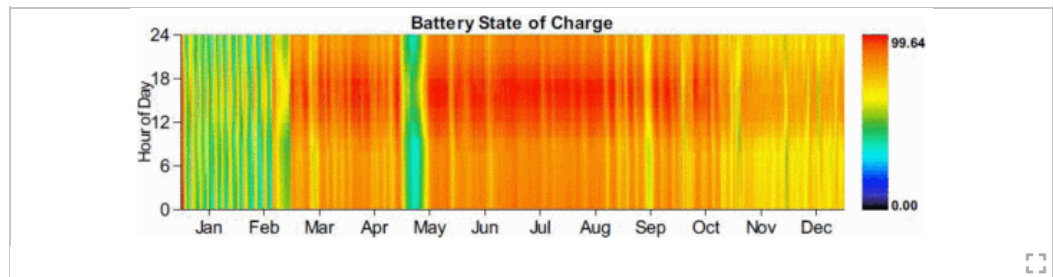


Figure 10.
Battery state of charge (%)

The electrical power generated by the sources is used to feed the school load. The average consumption of the primary load is 8450 kWh per year. The monthly average electricity production of the solar PV system is shown in Fig. 7. It can be seen that the total power production of the solar PV system is 6911 kWh per year, diesel generator power production is 4054 kWh per year with 2251 liters per year and the Annual throughput of the battery is 5327 kWh per year. Excess electricity is 650 kWh per year with zero percentage unmet electrical load.

•: Scenario 2 System architecture (Wind turbine/Generator/ Battery)

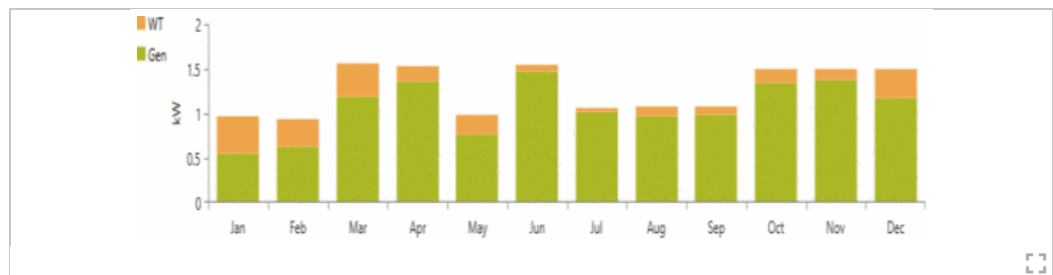


Figure 11.
Monthly average electric production (wind turbine/generator).

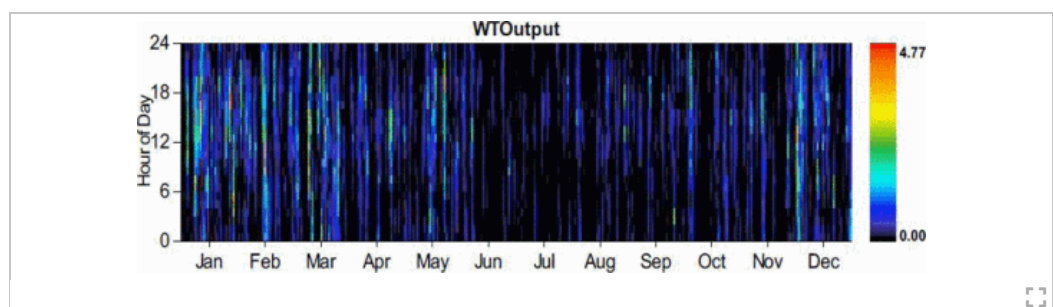


Figure 12.
Wind turbine power output (kw).

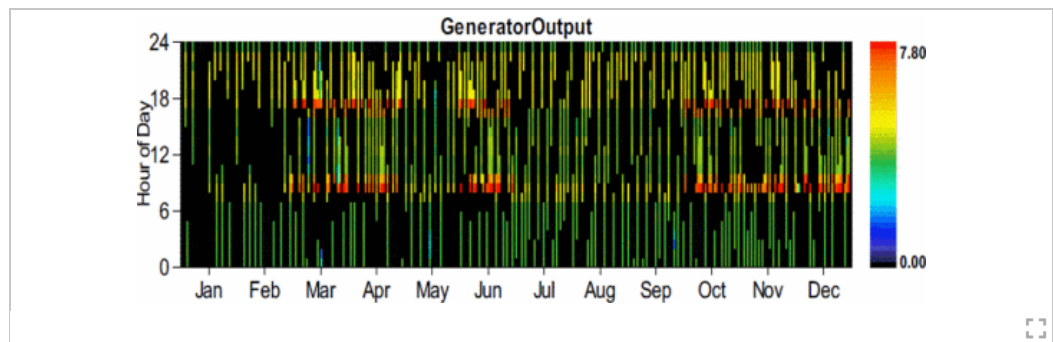


Figure 13.
Generator power output (kw).

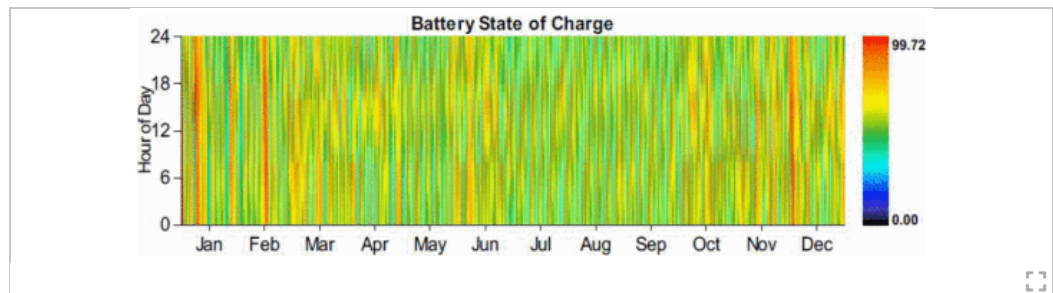


Figure 14.
Battery state of charge (%).

The electrical power generated by the sources is used to feed the school load. The average consumption of the primary load is 8450 kWh per year. The monthly average electricity production of the Wind turbine system is shown in [Fig. 11](#). It can be seen that the total power production of the wind turbine system is 1773 kWh per year; diesel generator power production is 9361 kWh per year with 5066 liters per year and the Annual throughput of the battery is 6144 kWh per year. Excess electricity is 9 kWh per year with zero percentage unmet electrical load.

•: Scenario 3 System architecture (PV /Wind turbine/Generator/ Battery)

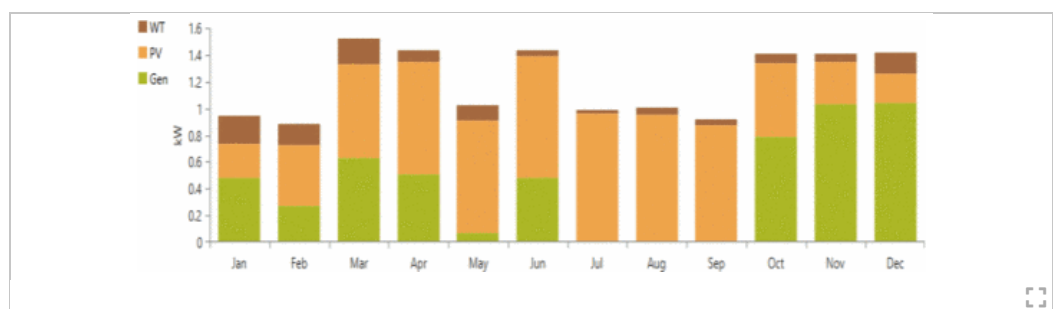


Figure 15.
Monthly average electric production (pv /wind turbine/generator).

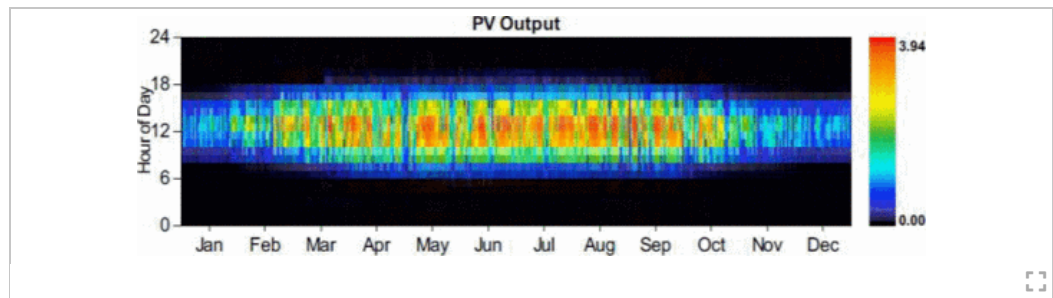


Figure 16.
PV power output (kw).

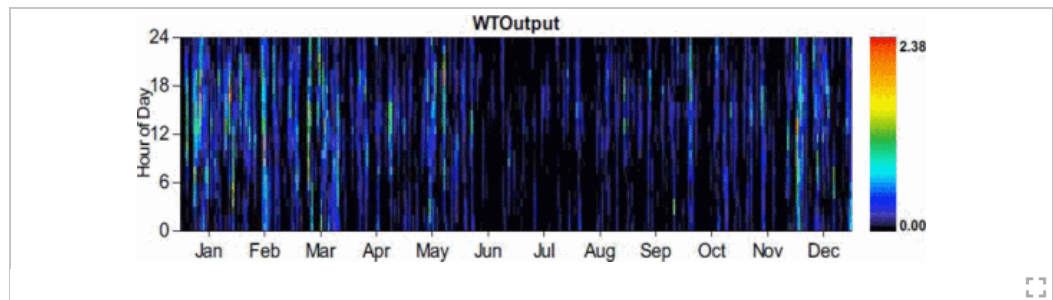


Figure 17.
Wind turbine power output (kw).

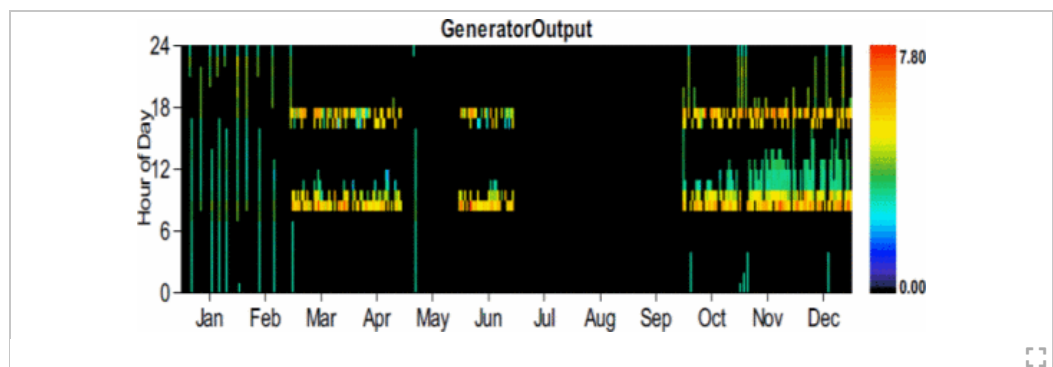


Figure 18.
Generator power output (kw).



Figure 19.
Battery state of charge (%)

The electrical power generated by the sources is used to feed the school load. The average consumption of the primary load is 8450 kWh per year. The monthly average electricity production of the Wind turbine system and PV solar system is shown in Fig. 15. It can be seen that the total power production of the wind turbine system is 886kWh per year, the total power production of PV system is 5759 kWh per year, diesel generator power production is 3869 kWh per year with 2139 liters per year and the Annual throughput of the battery is 5025 kWh per year. Excess electricity is 263 kWh per year with zero percentage unmet electrical load.

B. Economic Evaluation

Economic evaluation of the entire hybrid system is achieved by optimizing total life cycle cost of the system configurations. Total net present cost (NPC) is used to represent the system life-cycle cost, which is a

composite of initial system component's capital cost, replacement cost, operation and maintenance cost as well as fuel cost where applicable over their useful lifetime [14], [15].



Figure 20.

Economic comparison of various systems configurations.

The capital cost, replacement cost, O&M cost and total cost of the individual components both for one year and for a lifetime of 25 years have been shown in Fig. 20 from the software for the three scenarios.

The capital cost of the system is estimated at 55750 for the scenario 2 and 56550 \$ for the scenario 3. It is seen that the capital cost of diesel generator is higher than that of solar energy and wind energy for the three scenarios. So he represents a major share in the total capital cost of the system.

The operation and maintenance cost of the diesel generator is also relatively higher for the scenario 3 compared by the 1 and 2 scenarios. Hence it can be deduced that the inclusion of diesel generator in the system is the driving factor which decides the overall cost of the system. Also, it is seen that over the lifetime of the system, a considerable would be incurred in the form of replacement costs only in the case of battery replacement. The total net presents cost of the system considering the capital cost and other costs incurred over the lifetime of 25 years is estimated 122202 for the scenario 2 and 122336 for the scenario 3.

C. Emission Analysis

Table 4 shows the annual pollutant emission for the three configurations. This emission includes; carbon dioxide (CO₂), carbon monoxide (CO), unburned hydrocarbon (UHC), particulate matter (PM), sulfur dioxide (SO₂) and finally nitrogen oxide (NO_x). The conventional diesel based power generation in the first configuration produces 9927 kg/yr of CO₂, 13340 kg/year in the second configuration. While the optima hybrid renewable system configurations produced 5632 kg/year of CO₂. From this table, we observed that, a total of 74%, 77% CO₂ emission be prevented; if a hybrid renewable energy system third configurations is adopted. The percentage reduction in pollutants emission is proportional to the rate of renewable energy penetration in the system (see Fig. 21). Therefore, inclusion of renewable energy resources such as PV and wind in the hybrid system configuration will definitely lead to more economically viable option in providing electricity to rural school centers. Moreover, the system will also help in abating the prevalent global warming, which occurred due to high rate of CO₂ emission into the environment.



Figure 21.

(a) Frequency of changes in renewable penetration over 1 hr (pv / generator), (b) frequency of changes in renewable penetration over 1 hr (wind turbine/generator), (c) frequency of changes in renewable penetration over 1 hr (pv/wind turbine/generator)

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Table IV. Pollutant emission in optimal system configuration for the three proposed configurations	▼
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Conclusion	

The purpose of this study is to simulate using renewable energy as an alternative to meet the electricity needs for an elementary school in the rural environment and to inculcate to students the importance of renewable energy from an early age, besides a simulation using HOMER Pro software of the three architectures proposed composed of PV, wind turbine, battery and conventional diesel generator systems.

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Economic and environmental comparison for each configuration evaluated the best system configuration and is well proved through the results presented in this paper. Thus HOMER Pro is advantageous in performing energy balance configuration for each hour to choose feasible configuration.

Thus cost estimation of any feasible combination for installing and operating over the life time at a particular area can successfully be analyzed using the proposed methodology.

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